Algebra 1
Unit 3
Lesson 1 Practice

Name ___Answer Key___
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Period ____________________

Use the equation $f(x) = \frac{5}{3}x - 1$ for questions 1 and 2.

1. What is the form of this linear equation?

   This equation is in slope-intercept form.

2. Rewrite this equation in standard form.

$$y = \frac{5}{3}x - 1$$
$$\frac{-5}{3}x \qquad \frac{-5}{3}x$$

$\boxed{\frac{-5}{3}x + y = -1}$ or any equivalent equation in standard form.

Use the equation $y + 1 = -\frac{1}{3}(x - 3)$ for questions 3 and 4.

3. What is the form of this linear equation?

   This equation is in point-slope form.

4. Rewrite this equation in standard form.

$$y + 1 = \frac{-1}{3}(x - 3)$$
$$y + 1 = \frac{-1}{3}x + 1$$
$$+\frac{1}{3}x \qquad +\frac{1}{3}x$$
$$\frac{1}{3}x + y + 1 = 1$$

$\boxed{\frac{1}{3}x + y = 0}$ or any equivalent equation in standard form.

Use the equation $6x + 2y = -8$ for questions 5 and 6.

5. What is the form of this linear equation?

   This equation is in standard form.

6. Rewrite this equation in slope-intercept form.

$$6x + 2y = -8$$
$$-6x \qquad -6x$$
$$\frac{2y}{2} = \frac{-6x}{2} - \frac{8}{2}$$
$$\boxed{y = -3x - 4}$$

Use the equation $y+3=-2(x+7)$ for questions 7 through 10.

7. What is the slope of the line?

The slope is -2 $\quad y-y_1=m(x-x_1)$

8. Rewrite this equation in slope-intercept form.

$$\begin{aligned} y+3 &= -2(x+7) \\ y+3 &= -2x-14 \\ -3 \quad & \quad\quad\quad -3 \end{aligned}$$

$$\boxed{y = \underset{\substack{\downarrow \\ \text{m}}}{-2}x \underset{\substack{\downarrow \\ \text{b}}}{-17}} \qquad y=mx+b$$

9. Identify the y-intercept of the line.

The y-intercept is $(0,-17)$

10. Rewrite the equation in standard form.

$$\begin{aligned} y &= -2x-17 \\ +2x & \quad\quad +2x \end{aligned}$$

$2x+y=-17$ or any equivalent equation in standard form.

Algebra 1
Unit 3
Lesson 2 Practice

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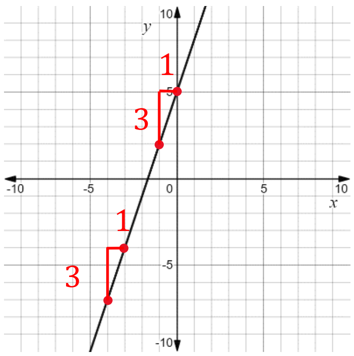
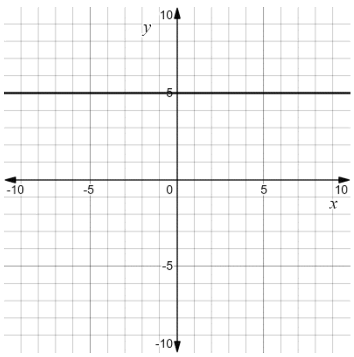
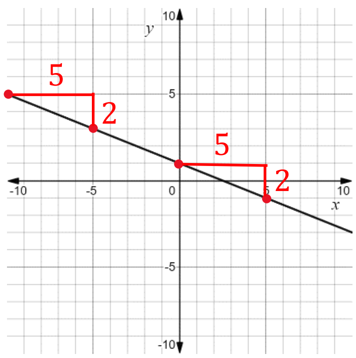
For questions 1 through 4, determine the slope of the line shown.

1. 

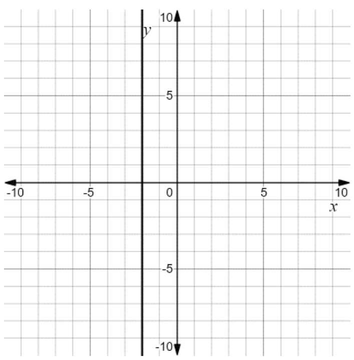
$m =$ $\frac{3}{1} = 3$

2. 

$m =$ 0

3. 

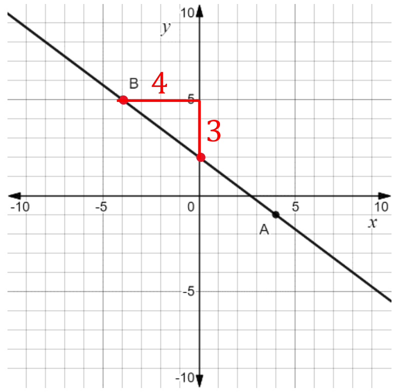
$m =$ $-\frac{2}{5}$

4. 

$m =$ undefined

For questions 5 through 8, the graph of a line with points A and B labeled is shown.

$m = -\frac{3}{4}$



point $B = (-4,5)$

point $A = (4,-1)$

5. Write the equation of the line in point-slope form using point A.

$y + 1 = -\frac{3}{4}(x - 4)$

6. Write the equation of the line in point-slope form using point B.

$y - 5 = -\frac{3}{4}(x + 4)$

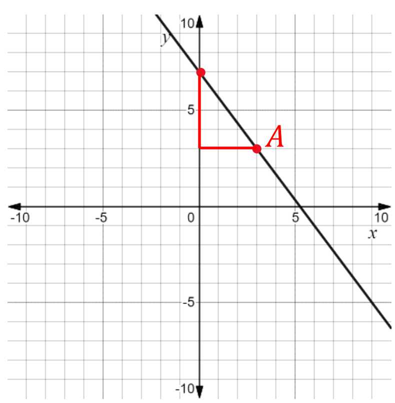
7. Rewrite the equation in point-slope form from point A in slope-intercept form.

$y + 1 = -\frac{3}{4}x + 3 \quad \rightarrow \quad y = -\frac{3}{4}x + 2$

8. Rewrite the equation in point-slope form from point B in standard form.

$4y - 20 = -3(x + 4) \quad \rightarrow \quad 4y - 20 = -3x - 12 \quad \rightarrow \quad 3x + 4y = 8$

For questions 9 and 10, the graph of a line is shown.



$m = -\frac{4}{3}$

point $A = (3,3)$

9. Write the equation of the line in point-slope form.

$y - 3 = -\frac{4}{3}(x - 3)$

10. Rewrite the equation from question 9 in standard form.

$y - 3 = -\frac{4}{3}x + 4 \quad \rightarrow \quad \frac{4}{3}x + y - 3 = 4 \quad \rightarrow \quad \frac{4}{3}x + y = 7$ or $4x + 3y = 21$

Algebra 1
Unit 3
Lesson 3 Practice

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For questions 1 through 4, a table of values is shown.

x	-2	0	3	7
y	-2.2	-3	-4.2	-5.8

1. Find the slope of the line. Show your work.
$m = \frac{-5.8-(-4.2)}{7-3} = \frac{-1.6}{4} = -0.4$ or $-\frac{4}{10}$ or $-\frac{2}{5}$

2. Write the equation of the line modeling this table in point-slope form.
Answers may vary. $y + 5.8 = -0.4(x - 7)$

3. Rewrite the equation of the line in slope-intercept form.
$y + 5.8 = -0.4x + 2.8 \quad \rightarrow \quad y = -0.4x - 3$

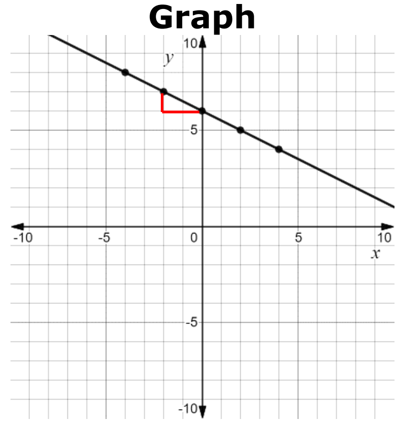
4. Rewrite the equation of the line in standard form.
$0.4x + y = -3 \quad \rightarrow \quad 4x + 10y = -30$

A table and graph for a linear function, $g(x)$ are shown. Use the table and graph for questions 5 and 6.

Table

x	g(x)
-4	8
-2	7
2	5

Graph

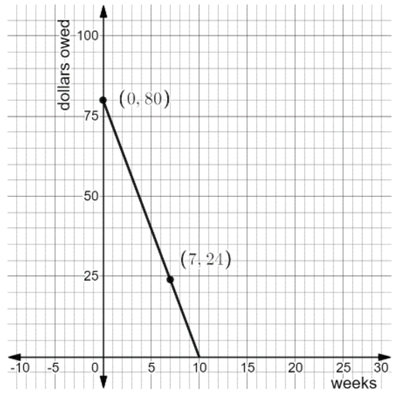


$m = -\frac{1}{2}$

5. Write the equation of the line in point-slope form.
Answers may vary. $y - 8 = -\frac{1}{2}(x + 4)$

6. Write the equation of the line in slope-intercept form.
$y - 8 = -\frac{1}{2}x - 2 \quad \rightarrow \quad y = -\frac{1}{2}x + 6$

Maria borrowed $\$80$ from her grandmother for summer camp. She agrees to repay her $\$8$ per week until he has paid the money back. The graph of this linear relationship is shown. Use the graph to answer questions 7 through 10.



7. Find the slope of the line.

$$m = \frac{24 - 80}{7 - 0} = \frac{-56}{7} = -8$$

8. Write the equation of the line in point slope form.

Answers may vary. $y - 24 = -8(x - 7)$

9. Write the equation of the line in slope-intercept form.

$y - 24 = -8x + 56$

$y = -8x + 80$

10. Write the equation of the line in standard form.

$8x + y = 80$

Algebra 1
Unit 3
Lesson 4 Practice

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A line passes through the points $(-8, 1)$ and $(1, 4)$. Use this information for questions 1 through 3.
(x_1, y_1) (x_2, y_2)

1. Find the slope of the line. Show your work.

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 1}{1 - (-8)} = \frac{3}{9} = \frac{1}{3}$$

2. Write the equation of the line modeling this table in point-slope form.

Answers may vary. $y - 1 = \frac{1}{3}(x + 8)$

3. Rewrite the equation of the line in standard form.

$$y - 1 = \frac{x}{3} + \frac{8}{3}$$

$$3(-\frac{x}{3} + y = \frac{8}{3} + 1)$$

$$-x + 3y = 8 + 3$$

$$-x + 3y = 11$$

Patty uses reclaimed rainwater in her garden. At 3 p.m., the water level in her rain barrel was 1.8 inches. During a rain shower, from 3 p.m. to 8 p.m., the rain fell at a steady rate. At 8 p.m., the water level in her rain barrel was 3.9. Use this information for questions 4 through 7.

4. Determine two ordered pairs on the graph of this relationship.

$(3, 1.8)$ and $(8, 3.9)$

5. Use the two ordered pairs to find the slope of the line.

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3.9 - 1.8}{8 - 3} = \frac{2.1}{5} = 0.42$$

6. Use the ordered pairs to write the equation of the line in point-slope form.

Answers may vary. $y - 1.8 = 0.42(x - 3)$

7. Use the two ordered pairs to write the equation of the line in slope-intercept form.

$$y - 1.8 = 0.42x - 1.26$$

$$y = 0.42x - 1.26 + 1.8$$

$$y = 0.42x + 0.54$$

A line has an x-intercept of 6 and a y-intercept of -2. Use this information for questions 8 through 10.

8. Find the slope of the line.

$(6, 0) \qquad (0, -2)$

$$\frac{-2-0}{0-6} = \frac{-2}{-6} = \frac{1}{3}$$

9. Write the equation of the line in slope-intercept form.

$$y = \frac{1}{3}x - 2$$

10. Write the equation of the line in standard form.

$$\left(-\frac{1}{3}x + y = -2\right)3$$

$$-x + 3y = -6$$

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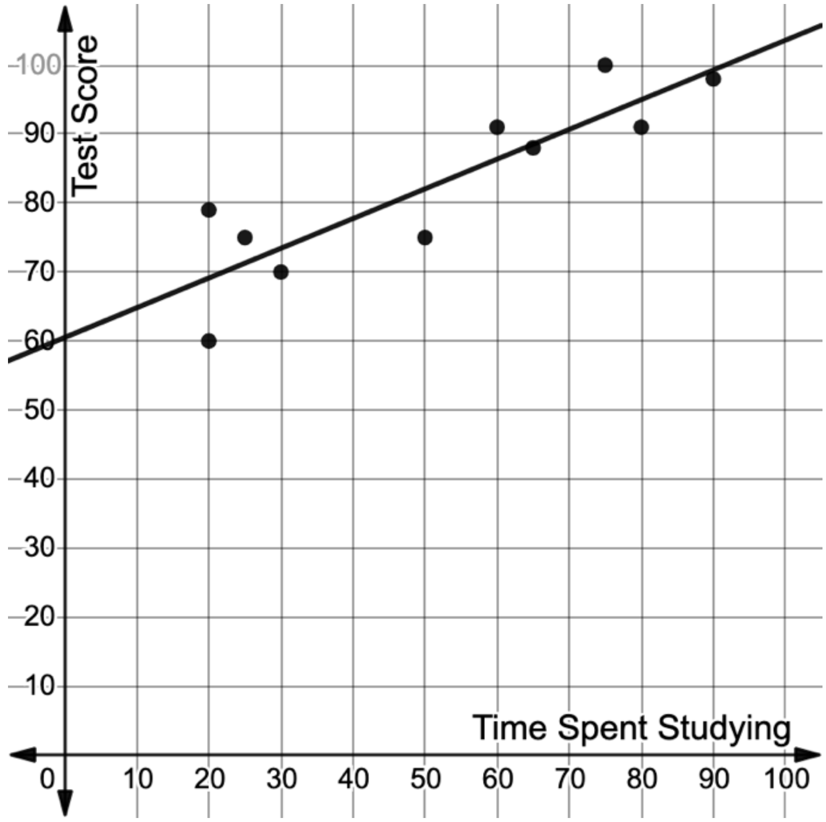
For questions 1 through 5, the scatter plot shows the relationship between the time students spent studying for a test, in minutes, and the test score earned.



The equation $y = 0.44x + 60.3$ can be used to model the relationship between the time spent studying, x, and the test score earned, y.

1. Find the slope of the model.

$m = 0.44$

2. Determine the y-intercept of the model.

60.3 or $(0, 60.3)$

3. What does the slope represent in the context?

Test scores increase 0.44 points for each minute students spend studying for a test.

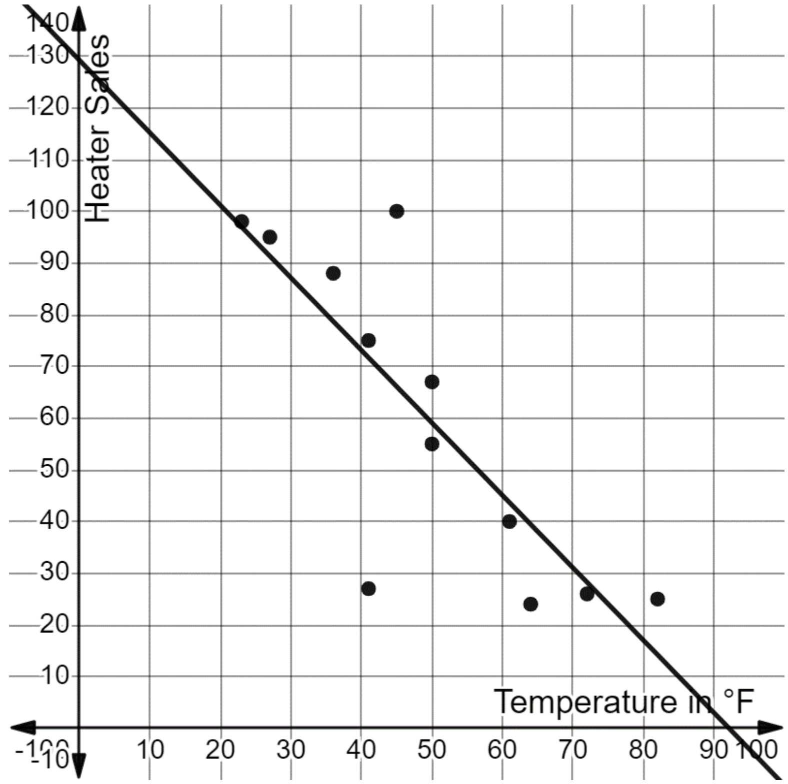
4. Determine an appropriate domain in the context.

$x \geq 0$

5. Determine an appropriate range in the context.

$y \geq 0$

For questions 6 through 10, the scatter plot shows the relationship between the average monthly temperature, in degrees Fahrenheit (°F), and the monthly sales of heaters.



The equation $y = -1.4x + 129.18$ can be used to model the relationship between the temperature, x, and heater sales, y.

6. Find the slope of the model.

$m = -1.4$

7. Determine the y-intercept of the model.

129.18 or $(0, 129.18)$

8. What does the slope represent in the context?

Sales decrease by 1.4 heaters as temperature increases by 1 degree (sales based on temperature).

9. Determine an appropriate domain in the context.

Answers may vary.
$x \geq -10$

10. Determine an appropriate range in the context.

Answers may vary.
$0 \leq y < 150$

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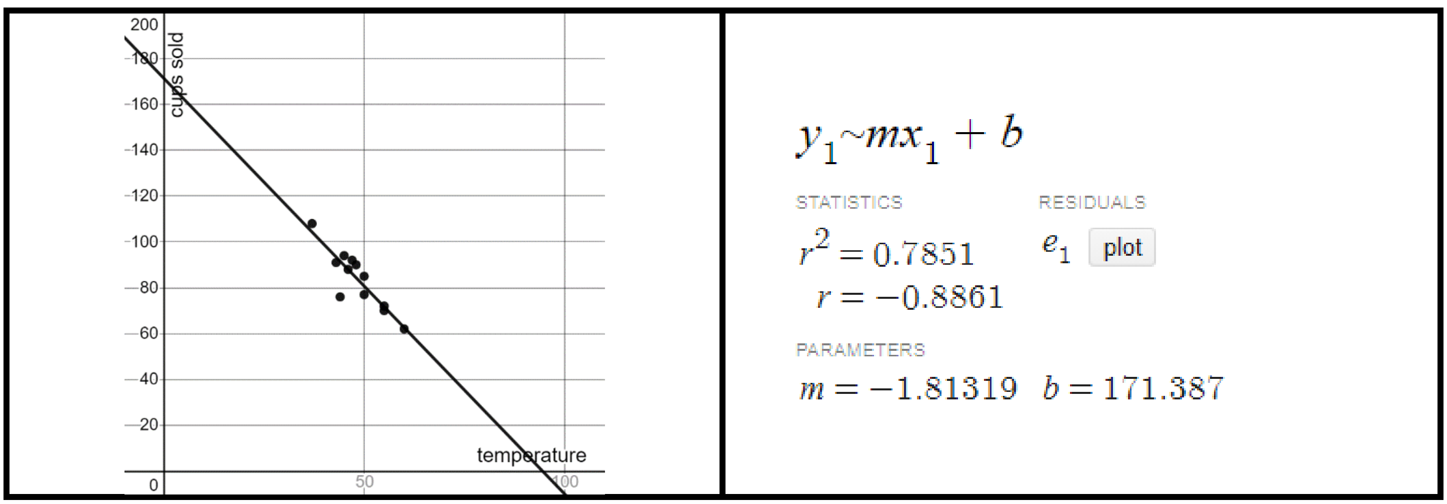
Use this scenario for questions 1 through 6. Polly sells hot chocolate at a stand in a park. She records the number of cups of hot chocolate she sells, as well as the average daily low temperature. The table shown indicates the data for 12 days in January.

Hot Chocolate Sales

Average Temperature (°F)	55	60	55	50	46	37	43	45	47	44	50	48
Cups of Hot Chocolate Sold	70	62	72	85	88	108	91	95	92	76	77	90

Polly used technology to generate a scatter plot and a line of best fit, as shown.



1. Write the equation for the line of best fit using the values generated with technology.

   $y = -1.81319x + 171.387$

2. Interpret the slope of the line of best fit in the context.

   As the temperature increases, the number of cups of hot chocolate sold decreases.

3. Describe the strength of the association between temperature and the number of cups of hot chocolate Polly sells.

   Strong

4. Describe the direction of the linear association between temperature and the number of cups of hot chocolate Polly sells.

The direction is negative.

5. Suppose Polly knows that tomorrow's temperature is predicted to be $32°\text{F}$. Based on the line of best fit, how many cups of hot chocolate should she anticipate selling?

$y = -1.81319(32) + 171.387$
$y = 113$ cups

6. A few weeks ago, Polly sold 104 cups of hot chocolate. Based on the line of best fit, what was the average temperature that day?

$104 = -1.81319x + 171.387$
$x = 37°$

A scatter plot created with technology generated the following values, as shown. Use the values for questions 7 through 10.

$y_1 \sim mx_1 + b$

STATISTICS

$r^2 = 0.8822$
$r = 0.9393$

RESIDUALS

e_1 plot

PARAMETERS

$m = 1.45686$ $b = -0.777333$

7. Describe the strength of the linear correlation for this data.

strong

8. Describe the direction of the linear correlation of this data.

positive

9. Use this data to write the equation of the line of best fit in slope-intercept form.

$y = 1.45686x - 0.777333$

10. Use the line of best fit to predict a value for y if the value of x is 9.

$y = 1.45686(9) - 0.777333$
$y = 12.334407$

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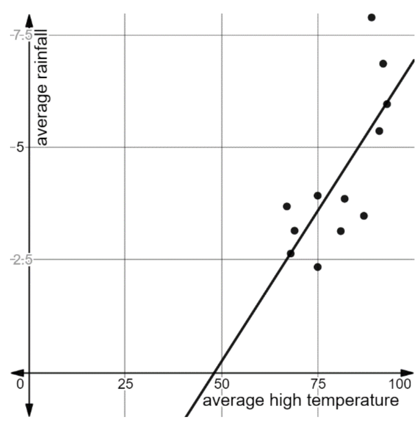
A table shows the average high temperature and the average rainfall each month for a year in Jacksonville, Florida.

Month	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec
Average High Temperature (°F)	67	69	75	81	87	91	93	92	89	82	75	68
Average Rainfall (in)	3.69	3.15	3.93	3.14	3.48	5.37	5.97	6.87	7.9	3.86	2.34	2.64

The scatter plot shown represents the data from the table with a linear equation fit to model the equation.



This screenshot shows the slope, y-intercept, and correlation coefficient associated with the linear relationship modeled on the scatter plot.

$$y_1 \sim mx_1 + b$$

STATISTICS　　RESIDUALS

$r^2 = 0.5537$　　e_1 plot

$r = 0.7441$

PARAMETERS

$m = 0.134323$　　$b = -6.48488$

1. Write the equation for the line of best fit shown.

$y = 0.134323x - 6.48488$

2. Interpret the meaning of the slope in terms of the context.

   As the average high temperature increases, the average rainfall increases.

3. Interpret the meaning of the y-intercept in terms of the context.

   When the high temp is $0°$, the average rainfall is -6.48488 which doesn't make sense in this context.

4. Interpret the meaning of the correlation coefficient in terms of the context.

   Strong, Positive, Linear

5. Use an inequality to express an appropriate domain in terms of the context.

   Answers will vary. $50 \leq x \leq 100$

6. Use an inequality to express an appropriate range in terms of the context.

   Answers will vary. $0 \leq y \leq 10$

7. Use the model to predict the average rainfall in a month with an average high temperature of $97°\text{F}$.

   $y = 0.134323\,(97) - 6.48488$
   $y = 6.54$ inches

8. Use the model to predict the average rainfall in a month with an average high temperature of $55°\text{F}$.

   $y = 0.134323\,(55) - 6.48488$
   $y = 0.90$ inches

9. Use the model to predict the average temperature in a month with an average rainfall of 3 inches.

   $3 = 0.134323x - 6.48488$
   $x = 70.6°$

10. Use the model to predict the average temperature in a month with an average rainfall of 5 inches.

    $5 = 0.134323x - 6.48488$
    $x = 85.5°$

Algebra 1
Unit 3
Lesson 8 Practice

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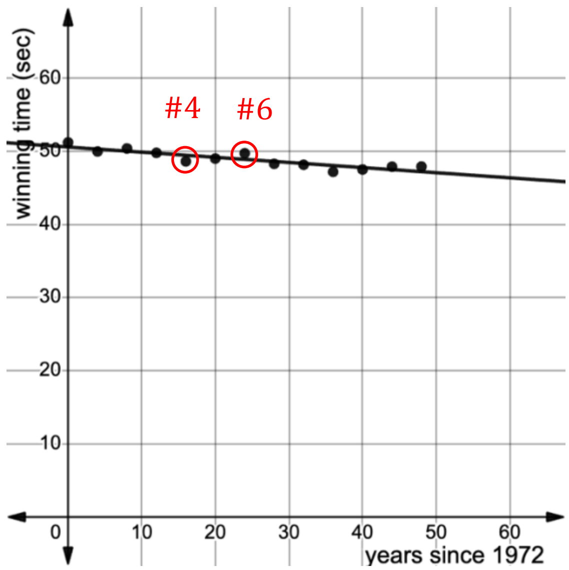
A scatter plot representing the gold medal winning times for the mens' 100-meter freestyle swimming race since 1972 is shown.



The raw data used to construct the scatter plot is shown.

Year	**1972**	**1976**	**1980**	**1984**	**1988**	**1992**	**1996**
Winning Time (in seconds)	51.22	49.99	50.4	49.8	48.63	49.02	49.74
Year	**2000**	**2004**	**2008**	**2012**	**2016**	**2020**	
Winning Time (in seconds)	48.3	48.17	47.21	47.52	47.9	47.92	

The equation of the line of best fit that models this data is $f(x) = -0.07x + 50.597$, where f is the winning time of the mens' 100-meter freestyle even, and x is the number of years since 1972.

1. How many years after 1972 was the 2012 Olympics?
   40 years

2. Using the equation of the line of best fit, find the predicted finishing time of the gold medalist in 2012.
   $f(x) = -0.07(40) + 50.597$
   $f(x) = 47.8$ seconds

3. Identify the residual value of the point representing the 2012 winning time.
   $47.52 - 47.8 = -0.28$

4. Identify one point on the scatter plot where the predicted value is above the actual value.

$$1988 - \ (16{,}48.63)$$

5. Find the residual value of the point selected for question 4.

$$48.63 - 49.5 = -0.87$$

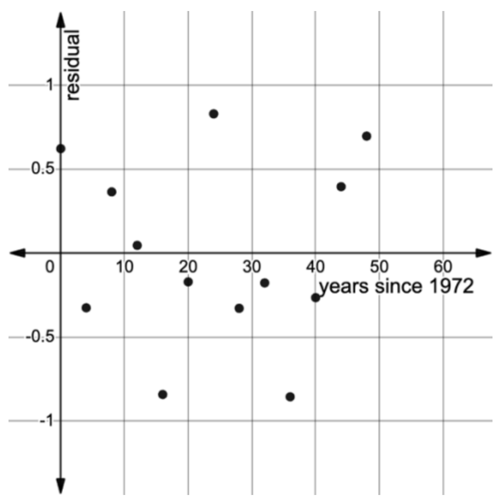
6. Identify one point on the scatter plot where the predicated value is below the actual value.

$$1996\ (24{,}49.74)$$

7. Find the residual value of the point selected for question 6.
$49.74 - 48.917 = 0.823$

8. Does the data appear to have a strong linear correlation? Explain how you know.
Because of the nearness of the data to the line of fit, it's a strong linear correlation.

The residual plot for the scatter plot is shown.



9. Explain how the residual values support the answer from question 8.

There does not seem to be any pattern which means there is a linear correlation.